

Appendix

APPENDIX A: LINEARIZATION OF (17)

First, a nonnegative auxiliary variable $\Delta P_{t,n}^x$ is introduced to characterize the deviation of the adjustable load power from its original demand, namely:

$$\Delta_{t,n,x} = |P_{t,n,x} - P_{t,n,x}^*|,$$

based on the equivalent representation of the absolute value, we have:

$$|P_{t,n,x} - P_{t,n,x}^*| = \max \{ (P_{t,n,x} - P_{t,n,x}^*), -(P_{t,n,x} - P_{t,n,x}^*) \}$$

the above relationship can be characterized by the following linear inequalities:

$$\begin{cases} \Delta_{t,n,x} \geq 0, \\ \Delta_{t,n,x} \geq P_{t,n,x} - P_{t,n,x}^*, \\ \Delta_{t,n,x} \geq -(P_{t,n,x} - P_{t,n,x}^*). \end{cases}$$

Accordingly, equation (17) in the manuscript can be equivalently rewritten in the following linear form:

$$C_{t,n,loa.} = \sum_{x \in \mathcal{X}_n} (1 - \mathbb{I}_{x,loa.})(\lambda_{n,x} \Delta_{t,n,x}),$$

together with the above constraint conditions.

APPENDIX B: POWER FLOW CONSTRAINTS (20)-(25)

For: $\forall n \in \mathcal{N}, t \in \mathcal{T}, \forall mn \in \mathcal{L}$,

$$\sum_{h \in \mathbb{E}(n)} P_{t,nh} = \sum_{m \in \mathbb{S}(n)} (P_{t,mn} - r_{mn} c_{t,mn}) + \sum_{x \in \mathcal{X}_n} P_{t,n,x} + (P_{t,n,b} - P_{t,n,f}), \quad (20)$$

$$\sum_{h \in \mathbb{E}(n)} Q_{t,nh} = \sum_{m \in \mathbb{S}(n)} (Q_{t,mn} - x_{mn} c_{t,mn}) + \sum_{x \in \mathcal{X}_n} Q_{t,n,x}, \quad (21)$$

$$\begin{aligned} v_{t,m} - v_{t,n} &= 2(r_{mn} P_{t,mn} + x_{mn} Q_{t,mn}) \\ &\quad - (r_{mn}^2 + x_{mn}^2) c_{t,mn}, \end{aligned} \quad (22)$$

$$c_{t,mn} = \frac{P_{t,mn}^2 + Q_{t,mn}^2}{v_{t,m}}, \quad (23)$$

$$\bar{v}_n \leq v_{t,n} \leq \underline{v}_n, \quad (24)$$

$$0 \leq c_{t,mn} \leq \bar{c}_{mn}, \quad (25)$$

(20) and (21) represent the active and reactive power balance constraints at node n in period t , respectively. Here, $P_{t,nh}$, $Q_{t,nh}$, $P_{t,mn}$, and $Q_{t,mn}$ denote the active and reactive power flows on branches nh and mn . r_{mn} and x_{mn} are the impedance parameters of branch mn , and $c_{t,mn}$ denotes the squared current of branch mn in period t . The sets $\mathbb{S}(n)$ and $\mathbb{E}(n)$ denote the upstream node set, e.g., $m \in \mathbb{S}(n)$, and the downstream node set, e.g., $h \in \mathbb{E}(n)$, of node n , respectively. (22) gives the network power flow equation, where $v_{t,m}$ and $v_{t,n}$ are the squared voltages of nodes m and n , respectively, and $\mathcal{L}$ is the set of distribution network

branches. (23) represents Ohm's law for branches and is convexified using second-order cone relaxation. (24) and (25) impose the node voltage and branch current limits, where $\bar{v}_n$ and $\underline{v}_n$ are the upper and lower voltage limits of node n , respectively, and $\bar{c}_{mn}$ is the current limit of branch mn .

APPENDIX C: LINEARIZATION OF (51) AND (54)

The frequency regulation tracking error in (51) is defined in the form of a continuous-time squared integral. To make this objective amenable to numerical solution, the interval $[0, \Delta\tau]$ is uniformly discretized into Z sampling points with step size $\omega = \Delta\tau/Z$, and the discrete time instants are defined as $s_z = z\omega, z = 1, 2, \dots, Z$. Accordingly, (51) can be rewritten as:

$$\vartheta_k \approx \sum_{z \in \mathcal{Z}} \left[R_t(k) - \sum_{n \in \mathcal{N}} \sum_{x \in \mathcal{X}_n} h_{t,n,x}(k, \tau_z) \right]^2,$$

(54) follows the same interpretation.

APPENDIX D: PROOF OF THE SUBLINEAR DYNAMIC REGRET BOUND

First, the problem is reformulated into a standard form, where the loss function at time period t is denoted by:

$$f(x_t) = - \sum_{n \in \mathcal{N}} [l_{t,n}(x_t) + \beta_{t,n} V(E_t(x_t))],$$

where $x_t \in X$ denotes the collection of all decision variables, including local decision variables and network coordination variables, and X is a closed, convex, and bounded set. In addition, all inequality constraints are denoted by $g(x_t) \leq 0$, while all equality constraints are denoted by $h(x_t) = 0$.

Define the per-period optimal feasible sequence as:

$$x_t \in \arg \min_{x \in X, g(x_t) \leq 0, h(x_t) = 0} f(x_t), \quad t = 1, 2, \dots, T.$$

Define the dynamic regret as:

$$DReg(T) := \sum_{t \in \mathcal{T}} [f(x_t) - f(x_t^*)].$$

Define the path variation of the comparator sequence as:

$$P_T := \sum_{t \in \mathcal{T}} \|x_t^* - x_{t-1}^*\|.$$

Define the cumulative soft-constraint violation as:

$$Vio_{soft}(T) := \left\| \sum_{t \in \mathcal{T}} [g(x_t)]_+ \right\|.$$

Assumption A1: Bounded feasible set

X is a nonempty, closed, convex, and compact set, and its diameter is bounded. That is, there exists $D > 0$ such that $\|x - y\| \leq D, \forall x, y \in X$.

Assumption A2: The loss function is convex and has bounded subgradients

Each f is convex with respect to x , and there exists a constant $G > 0$ such that $\|\nabla f(x_t)\| \leq G, \forall x \in X, \forall t$.

For this paper, this assumption holds because $\mathcal{L}_{t,n}$ consists of linear revenue terms and convex cost terms, while $V(E_t)$ is a quadratic term, and β_t is kept bounded through projection.

Assumption A3: The constraint functions are convex and have bounded jacobians

Each component $g_m(x_t)$ is a convex function, and there exist constants $H > 0$ and $B_g > 0$ such that $\|\partial g_m(x_t)\| \leq H, |g_m(x_t)| \leq B_g, \forall x \in X, \forall t, m$.

Assumption A4: A per-period optimal feasible solution exists

For any t , the set $\{x \in X : g(x_t) \leq 0, h(x_t) = 0\}$ is nonempty, therefore, x_t^* exists.

Assumption A5: Bounded composite gradient

Define the generalized gradient as:

$$z_t = \nabla f(x_t) + J(x_t)^T Q_t,$$

where $J(x_t)$ denotes the Jacobian of $g(x_t)$, and Q_t is the virtual queue. Assume that there exists a constant $C > 0$ such that $\|z_t\| \leq C, \forall t$.

1) Baseline proximal update:

Equation (34) in the manuscript is essentially an online proximal update consisting of the linearized current loss, the queue-penalty term, and a quadratic regularization term, and can therefore be written in the following unified form:

$$\tilde{x}_{t+1} = \Pi_X [x_t - \alpha (\nabla f(x_t) + J(x_t)^T Q_t)], \quad (1)$$

where $\Pi_X(\cdot)$ denotes the projection onto X , and $\alpha > 0$ is the stepsize.

2) Virtual queue update:

$$Q_{t+1} = [Q_t + \eta g(x_t)]_+,$$

where $\eta > 0$ is the queue stepsize.

3) Predictive correction:

Equation (35) specifies that the corrected solution is accepted only when the predicted gradient satisfies both the reliability condition and the sufficient descent condition; otherwise, the baseline solution is retained. Therefore, there exists a nonnegative improvement term $\Delta_{t+1} \geq 0$ such that the actual PDOCO decision x_{t+1} satisfies:

$$f(x_{t+1}) \leq f(\tilde{x}_{t+1}) - \Delta_{t+1},$$

when the predictive correction is not accepted, $\Delta_{t+1} = 0$, when it is accepted, $\Delta_{t+1} > 0$. Therefore, it suffices to first establish the regret upper bound for the baseline proximal-queue update, and then subtract $\sum_{t \in \mathcal{T}} \Delta_t$ to obtain the regret upper bound of PDOCO.

By the first-order optimality condition of the projection operator, for any $u \in X$, we have:

$$\langle z_t, x_t - u \rangle \leq \frac{\|x_t - u\|^2 - \|\tilde{x}_{t+1} - u\|^2}{2\alpha} + \frac{\alpha}{2} \|z_t\|^2. \quad (1)$$

Since f is a convex function,

$$f(x_t) - f(x_t^*) \leq \langle \nabla f(x_t), x_t - x_t^* \rangle, \quad (2)$$

moreover, since $z_t := \nabla f(x_t) + J(x_t)^T Q_t$, it follows that:

$$\langle \nabla f(x_t), x_t - x_t^* \rangle = \langle z_t, x_t - x_t^* \rangle - \langle Q_t, J(x_t)(x_t - x_t^*) \rangle. \quad (3)$$

By the convexity of the constraint function, we have $g(x_t^*) \geq g(x_t) + J(x_t)(x_t - x_t^*)$, which can be rearranged as $J(x_t)(x_t - x_t^*) \geq g(x_t) - g(x_t^*)$.

Since x_t^* is feasible, we have $g(x_t^*) \leq 0$ and $Q_t \geq 0$. Therefore:

$$-\langle Q_t, J(x_t)(x_t - x_t^*) \rangle \leq -\langle Q_t, g(x_t) \rangle. \quad (4)$$

Combining (1), (2), (3), and (4) yields:

$$f(x_t) - f(x_t^*) \leq \frac{\|x_t - x_t^*\|^2 - \|\tilde{x}_{t+1} - x_t^*\|^2}{2\alpha} + \frac{\alpha}{2} \|z_t\|^2 - \langle Q_t, g(x_t) \rangle, \quad (5)$$

further, by the non-deterioration property of the predictive correction, $f(x_{t+1}) \leq f(\tilde{x}_{t+1}) - \Delta_{t+1}$, the actual PDOCO decision satisfies:

$$f(x_t) - f(x_t^*) \leq \frac{\|x_t - x_t^*\|^2 - \|\tilde{x}_{t+1} - x_t^*\|^2}{2\alpha} + \frac{\alpha}{2} \|z_t\|^2 - \langle Q_t, g(x_t) \rangle - \Delta_t. \quad (6)$$

The main difficulty in (6) is that the comparator x_t^* varies with t , so it cannot be handled directly. To address this, we rewrite $\|\tilde{x}_{t+1} - x_t^*\|^2$ as $\|(\tilde{x}_{t+1} - x_{t+1}^*) + (x_{t+1}^* - x_t^*)\|^2$. Then, by using $\|a + b\|^2 \leq \|a\|^2 + 2\|a\|\|b\| + \|b\|^2$, together with $\|\tilde{x}_{t+1} - x_{t+1}^*\| \leq D$, we obtain:

$$-\|\tilde{x}_{t+1} - x_t^*\|^2 \leq -\|\tilde{x}_{t+1} - x_{t+1}^*\|^2 + 2D\|x_{t+1}^* - x_t^*\|, \quad (7)$$

substituting this into (6) yields:

$$f(x_t) - f(x_t^*) \leq \frac{\|x_t - x_t^*\|^2 - \|\tilde{x}_{t+1} - x_{t+1}^*\|^2}{2\alpha} + \frac{D}{\alpha} \|x_{t+1}^* - x_t^*\| + \frac{\alpha}{2} \|z_t\|^2 - \langle Q_t, g(x_t) \rangle - \Delta_t. \quad (8)$$

Summing both sides of (8) over t yields:

$$\sum_{t \in \mathcal{T}} [f(x_t) - f(x_t^*)] \leq \frac{\|x_1 - x_1^*\|^2}{2\alpha} + \frac{D}{\alpha} \sum_{t \in \mathcal{T}} \|x_{t+1}^* - x_t^*\| + \frac{\alpha}{2} \sum_{t \in \mathcal{T}} \|z_t\|^2 - \sum_{t \in \mathcal{T}} \langle Q_t, g(x_t) \rangle - \sum_{t \in \mathcal{T}} \Delta_t. \quad (9)$$

Since the diameter of X is bounded, we have $\|x_1 - x_1^*\|^2 \leq D^2$. Moreover, noting that $P_T = \sum_{t \in \mathcal{T}} \|x_{t+1}^* - x_t^*\|$, it follows that:

$$\text{DReg}(T) \leq \frac{D^2}{2\alpha} + \frac{D}{\alpha} P_T + \frac{\alpha}{2} \sum_{t \in \mathcal{T}} \|z_t\|^2 - \sum_{t \in \mathcal{T}} \langle Q_t, g(x_t) \rangle. \quad (10)$$

By A2-A3 and the boundedness analysis of the virtual queue, there exists a constant C such that $\|z_t\| \leq C, \forall t$. Thus:

$$\sum_{t \in \mathcal{T}} \|z_t\|^2 \leq C^2 T, \quad (11)$$

meanwhile, the last term, $-\sum_{t \in \mathcal{T}} \langle Q_t, g(x_t) \rangle$, is a non-positive gain term. Discarding it directly yields the following upper bound:

$$DReg(T) \leq \frac{D^2}{2\alpha} + \frac{D}{\alpha} P_T + \frac{\alpha C^2 T}{2}. \quad (12)$$

Let $\alpha = D/C\sqrt{T}$. Substituting it into (12) yields $DReg(T) \leq DC\sqrt{T} + CP_T\sqrt{T} - \sum_{t \in \mathcal{T}} \Delta_t$. Therefore:

$$DReg(T) \leq O\left(\sqrt{T}(1 + P_T)\right), \quad (13)$$

this completes the proof.

APPENDIX E: PROOF OF THE SUBLINEAR BOUND ON LONG-TERM SOFT-CONSTRAINT VIOLATION

Let $L_t := \frac{1}{2}\|Q_t\|^2$. From the queue update $Q_{t+1} = [Q_t + \eta g(x_t)]_+$, and by using $\|u_+\|^2 \leq \|u\|^2$, we obtain $\|Q_{t+1}\|^2 \leq \|Q_t + \eta g(x_t)\|^2 = \|Q_t\|^2 + 2\eta \langle Q_t, \eta g(x_t) \rangle + \eta^2 \|g(x_t)\|^2$. Therefore:

$$L_{t+1} - L_t \leq \eta \langle Q_t, g(x_t) \rangle + \frac{\eta^2}{2} \|g(x_t)\|^2. \quad (14)$$

By **Assumption A3**, $\|g(x_t)\| \leq B_g$. Therefore:

$$L_{t+1} - L_t \leq \eta \langle Q_t, g(x_t) \rangle + \frac{\eta^2 B_g^2}{2}. \quad (15)$$

By removing the non-positive improvement term $-\Delta_t$ from the one-step inequality (8) in the above dynamic regret proof, we obtain:

$$\begin{aligned} f(x_t) - f(x_t^*) + \langle Q_t, g(x_t) \rangle &\leq \frac{\|x_t - x_t^*\|^2 - \|\tilde{x}_{t+1} - x_{t+1}^*\|^2}{2\alpha} \\ &\quad + \frac{D}{\alpha} \|x_{t+1}^* - x_t^*\| + \frac{\alpha}{2} \|z_t\|^2. \end{aligned} \quad (16)$$

Multiplying (15) by $1/\eta$ and then adding it to (16) yields:

$$\begin{aligned} f(x_t) - f(x_t^*) + \frac{L_{t+1} - L_t}{\eta} &\leq \frac{\|x_t - x_t^*\|^2 - \|\tilde{x}_{t+1} - x_{t+1}^*\|^2}{2\alpha} \\ &\quad + \frac{D}{\alpha} \|x_{t+1}^* - x_t^*\| + \frac{\alpha}{2} \|z_t\|^2 + \frac{\eta B_g^2}{2}. \end{aligned} \quad (17)$$

Summing over $t = 1, 2, \dots, T$, we obtain:

$$DReg(T) + \frac{L_{T+1} - L_1}{\eta} \leq \frac{D^2}{2\alpha} + \frac{D}{\alpha} P_T + \frac{\alpha C^2 T}{2} + \frac{\eta B_g^2 T}{2}. \quad (18)$$

Since $L_1 \geq 0$, and in order to derive the $O(\sqrt{T})$ bound on cumulative soft-constraint violation, the online decision and the comparator are assumed to lie in the same instantaneous objective domain, it follows that $DReg(T) \geq 0$. Therefore:

$$\frac{L_{T+1}}{\eta} \leq \frac{D^2}{2\alpha} + \frac{D}{\alpha} P_T + \frac{\alpha C^2 T}{2} + \frac{\eta B_g^2 T}{2}, \quad (19)$$

that is:

$$L_{T+1} \leq \eta \left(\frac{D^2}{2\alpha} + \frac{D}{\alpha} P_T + \frac{\alpha C^2 T}{2} + \frac{\eta B_g^2 T}{2} \right). \quad (20)$$

Under **Assumptions A1-A4**, if the virtual queue is updated according to $Q_{t+1} = [Q_t + \eta g(x_t)]_+$ with $\alpha = \Theta(T^{-1/2})$ and $\eta = \Theta(T^{-1/2})$, then it holds that $L_{T+1} = O(1 + P_T)$. Therefore, $\|Q_{T+1}\| = \sqrt{2L_{T+1}} = O(\sqrt{1 + P_T})$.

From the queue recursion $Q_{t+1} = [Q_t + \eta g(x_t)]_+$, it can be shown recursively that $Q_{T+1} \succeq \left[\eta \sum_{t \in \mathcal{T}} g(x_t) \right]_+$, where $\succeq$ denotes componentwise inequality. Hence:

$$\left\| \left[\sum_{t \in \mathcal{T}} g(x_t) \right]_+ \right\| \leq \frac{\|Q_{T+1}\|}{\eta}. \quad (21)$$

substituting $\|Q_{T+1}\| = \sqrt{2L_{T+1}} = O(\sqrt{1 + P_T})$ and using $\eta = \Theta(T^{-1/2})$, we obtain:

$$Vio_{\text{soft}}(T) = \left\| \left[\sum_{t \in \mathcal{T}} g(x_t) \right]_+ \right\| \leq O\left(\sqrt{T}\sqrt{1 + P_T}\right), \quad (22)$$

this completes the proof.

CASE STUDIES

Simulation Setting

Renewable energy forecasting results on the test day and PJM market frequency regulation signal: The actual prices and their forecasts on the test day are shown in Fig. 1, while the actual renewable generation and load power, together with their forecasts, are shown in Fig. 2. Based on the actual operating data of the test day and the PJM frequency regulation signal, real-time dispatch and fast frequency response tests are subsequently conducted. The results show that the forecasting errors are generally controlled within 10%, providing reliable look-ahead information for online optimization. Since the forecasts are updated in a rolling manner, only the forecast results corresponding to each period are presented in the figures.

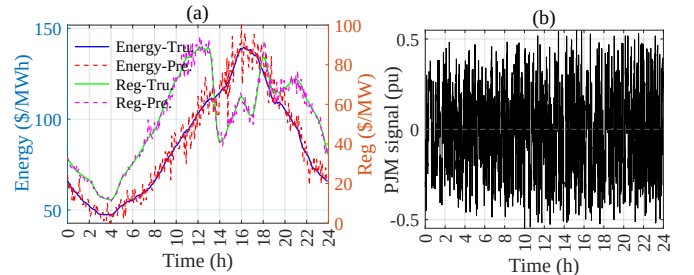


Fig. 1. Actual and forecast values of energy and reserve prices and PJM regulation signal: (a) actual and forecast values of energy and reserve prices. (b) PJM regulation signal.

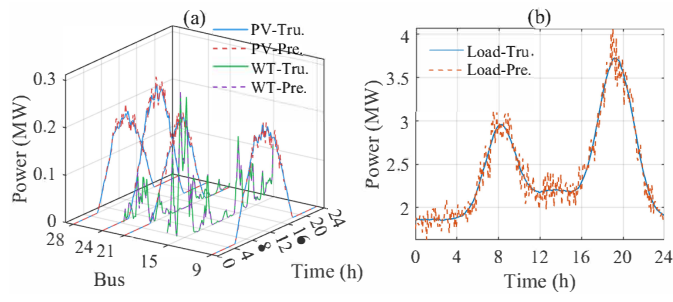


Fig. 2. Actual and forecast values of renewable generation and load power: (a) Actual and forecast values of renewable generation. (b) Actual and forecast values of load power.